ARISTA MAR-ACC-36 Site-Specific SOP for Critical Optical Transport Failure

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9206

Description

The MAR-ACC-36 site, being an Arista-enabled access node in the Marathahalli Core, is susceptible to critical optical transport failures stemming from SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This SOP outlines the procedures to diagnose and resolve such failures on this specific node, where the Arista CLI dialect is employed to navigate the control and data planes. Given the critical nature of these failures, it is essential to promptly isolate the root cause, execute the correct troubleshooting and resolution steps, and verify the efficacy of the fixes without compromising the network's integrity. The Arista-specific CLI commands used in this SOP are crucial in accessing and manipulating the underlying protocols, such as IS-IS, OSPF, and BGP, which govern the node's operation.

Pre-requisite Checks

- Ensure that remote hands are dispatched to the Marathahalli Core to facilitate physical access to the MAR-ACC-36 node.

- Verify the status of redundant links to determine if any alternative paths can be engaged to mitigate the impact of the failure.

- Confirm that all necessary safety protocols are in place before proceeding with any physical interventions.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to aging or manufacturing defects.	High
Fiber cuts or breaks within the optical link.	Medium
Rx/Tx power loss due to misaligned fiber connections or contamination.	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status | include MAR-ACC-36, Expected Output: MAR-ACC-36: up, link up (SFP/XFP status: unknown), Description: This command checks the interface status of MAR-ACC-36, providing insight into the link's operational state and the SFP/XFP's status.

Command: show log | grep ALARM_9206, Expected Output: ALARM_9206: Critical optical transport failure detected on MAR-ACC-36, Description: This command queries the system logs for any occurrences of the ALARM_9206 code, which is indicative of a critical optical transport failure.
```

```
Command: show interface detail | include MAR-ACC-36, Expected Output: MAR-ACC-36: optical receive power: 0 dBm, optical transmit power: 14 dBm, Description: This command provides a detailed view of the interface, including the optical receive and transmit power, which can help identify any issues related to Rx/Tx power loss.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: reload module 1, Description: This command initiates a reload of the SFP/XFP module, which should restore the link's operational state and resolve the failure.

Step 2: Step: Verify and reseat fiber connections, Command: verify fiber connections, Description: This step involves visually inspecting the fiber connections to ensure they are properly seated and not contaminated. If issues are found, reseat the fibers according to the manufacturer's instructions.

Step 3: Step: Perform a power cycle, Command: system power-cycle, Description: This command initiates a power cycle of the node, which can help resolve issues related to Rx/Tx power loss or other hardware-related failures.

Post-Check Verification

```
Command: show interface status | include MAR-ACC-36, Expected Output: MAR-ACC-36: up, link up (SFP/XFP status: operational), Description: This command verifies that the interface is operational and the SFP/XFP is functioning correctly.

Command: show log | grep ALARM_9206, Expected Output: ALARM_9206: Critical optical transport failure cleared, Description: This command checks if the ALARM_9206 code has been cleared, indicating that the failure has been resolved.

Command: show interface detail | include MAR-ACC-36, Expected Output: MAR-ACC-36: optical receive power: 10 dBm, optical transmit power: 14 dBm, Description: This command verifies that the optical receive and transmit power are within the expected ranges, indicating a successful resolution of the failure.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: reload module 1, Description: If the fix fails, this step involves reloading the original SFP/XFP module to revert to the previous configuration.

- Step: Re-verify fiber connections, Command: verify fiber connections, Description: This step involves re-inspecting the fiber connections to ensure they are properly seated and not contaminated. If issues are found, reseat the fibers according to the manufacturer's instructions.